

The gcloud CLI cheat sheet

A roster of go-to commands for the Google Cloud CLI, the primary command-line tool for Google Cloud.

(Also included: [introductory primer](#) (#introductory_primer), [understanding commands](#) (#understanding_commands), and a  [printable PDF](#) (/static/sdk/docs/images/gcloud-cheat-sheet.pdf).)

Cheat sheet

Getting started

Get going with the gcloud CLI.

- **[gcloud init](#)** (/sdk/gcloud/reference/init): Initialize, authorize, and configure the gcloud CLI.
- **[gcloud version](#)** (/sdk/gcloud/reference/version): Display version and installed components.
- **[gcloud components install](#)** (/sdk/gcloud/reference/components/install): Install specific components.
- **[gcloud components update](#)** (/sdk/gcloud/reference/components/update): Update your gcloud CLI to the latest version.
- **[gcloud config set project](#)** (/sdk/gcloud/reference/config/set): Set a default Google Cloud project to work on.
- **[gcloud info](#)** (/sdk/gcloud/reference/info): Display current gcloud CLI environment details.

Help

gcloud CLI is happy to help.

- **[gcloud help](#)** (/sdk/gcloud/reference/help): Search the gcloud CLI reference documents for specific terms.

- **gcloud feedback** (/sdk/gcloud/reference/feedback): Provide feedback to the gcloud CLI team.
- **gcloud topic** (/sdk/gcloud/reference/topic): Supplementary help material for non-command topics like accessibility, filtering, and formatting.

Personalization

Make the gcloud CLI your own; personalize your configuration with properties.

- **gcloud config set** (/sdk/gcloud/reference/config/set): Define a property (like compute/zone) for the current configuration.
- **gcloud config get** (/sdk/gcloud/reference/config/get): Fetch the value of a gcloud CLI property.
- **gcloud config list** (/sdk/gcloud/reference/config/list): Display all the properties for the current configuration.
- **gcloud config configurations create**
(/sdk/gcloud/reference/config/configurations/create): Create a new named configuration. 
- **gcloud config configurations list** (/sdk/gcloud/reference/config/configurations/list): Display a list of all available configurations.
- **gcloud config configurations activate**
(/sdk/gcloud/reference/config/configurations/activate): Switch to an existing named configuration.

Authorization and Credentials

Grant and revoke authorization to the gcloud CLI and manage credentials.

- **gcloud auth login** (/sdk/gcloud/reference/auth/login): Authorize Google Cloud access for the gcloud CLI with Google Cloud user credentials and set the current account as active.
- **gcloud auth activate-service-account**
(/sdk/gcloud/reference/auth/activate-service-account): Authorize Google Cloud access similar to **gcloud auth login** but with service account credentials.
- **gcloud auth application-default** (/sdk/gcloud/reference/auth/application-default): Manage your Application Default Credentials (ADC) for Cloud Client Libraries.

- **gcloud auth list** (/sdk/gcloud/reference/auth/list): List all credentialed accounts.
- **gcloud auth print-access-token** (/sdk/gcloud/reference/auth/print-access-token): Display the current account's access token.
- **gcloud auth revoke** (/sdk/gcloud/reference/auth/revoke): Remove access credentials for an account.

Projects

Manage project access policies.

- **gcloud projects describe** (/sdk/gcloud/reference/projects/describe): Display metadata for a project (including its ID).
- **gcloud projects add-iam-policy-binding** (/sdk/gcloud/reference/projects/add-iam-policy-binding): Add an IAM policy binding to a specified project.

IAM



Configuring Identity and Access Management (IAM) preferences and service accounts.

- **gcloud iam list-grantable-roles** (/sdk/gcloud/reference/iam/list-grantable-roles): List IAM grantable roles for a resource.
- **gcloud iam roles create** (/sdk/gcloud/reference/iam/roles/create): Create a custom role for a project or org.
- **gcloud iam service-accounts create** (/sdk/gcloud/reference/iam/service-accounts/create): Create a service account for a project.
- **gcloud iam service-accounts add-iam-policy-binding** (/sdk/gcloud/reference/iam/service-accounts/add-iam-policy-binding): Add an IAM policy binding to a service account.
- **gcloud iam service-accounts set-iam-policy-binding** (/sdk/gcloud/reference/iam/service-accounts/set-iam-policy): Replace existing IAM policy binding.
- **gcloud iam service-accounts keys list** (/sdk/gcloud/reference/iam/service-accounts/keys/list): List a service account's keys.

Docker & Google Kubernetes Engine (GKE)

Manage containerized applications on Kubernetes.

- **gcloud auth configure-docker** (/sdk/gcloud/reference/auth/configure-docker): Register the gcloud CLI as a Docker credential helper.
- **gcloud container clusters create** (/sdk/gcloud/reference/container/clusters/create): Create a cluster to run GKE containers.
- **gcloud container clusters list** (/sdk/gcloud/reference/container/clusters/list): List clusters for running GKE containers.
- **gcloud container clusters get-credentials** (/sdk/gcloud/reference/container/clusters/get-credentials): Update kubeconfig to get kubectl to use a GKE cluster.
- **gcloud container images list-tags** (/sdk/gcloud/reference/container/images/list-tags): List tag and digest metadata for a container image.



Virtual Machines & Compute Engine

Create, run, and manage VMs on Google Cloud infrastructure.

- **gcloud compute zones list** (/sdk/gcloud/reference/compute/zones/list): List Compute Engine zones.
- **gcloud compute instances create** (/sdk/gcloud/reference/compute/instances/create): Create a VM instance.
- **gcloud compute instances describe** (/sdk/gcloud/reference/compute/instances/describe): Display a VM instance's details.
- **gcloud compute instances list** (/sdk/gcloud/reference/compute/instances/list): List all VM instances in a project.
- **gcloud compute disks snapshot** (/sdk/gcloud/reference/compute/disks/snapshot): Create snapshot of persistent disks.
- **gcloud compute snapshots describe** (/sdk/gcloud/reference/compute/snapshots/describe): Display a snapshot's details.

- **`gcloud compute snapshots delete`** (/sdk/gcloud/reference/compute/snapshots/delete): Delete a snapshot.
- **`gcloud compute ssh`** (/sdk/gcloud/reference/compute/ssh): Connect to a VM instance by using SSH.

Serverless & App Engine

Build highly scalable applications on a fully managed serverless platform

- **`gcloud app deploy`** (/sdk/gcloud/reference/app/deploy): Deploy your app's code and configuration to the App Engine server.
- **`gcloud app versions list`** (/sdk/gcloud/reference/app/versions/list): List all versions of all services deployed to the App Engine server.
- **`gcloud app browse`** (/sdk/gcloud/reference/app/browse): Open the current app in a web browser.
- **`gcloud app create`** (/sdk/gcloud/reference/app/create): Create an App Engine app within your current project.
- **`gcloud app logs read`** (/sdk/gcloud/reference/app/logs/read): Display the latest App Engine app logs.



Miscellaneous

Commands that might come in handy

- **`gcloud kms decrypt`** (/sdk/gcloud/reference/kms/decrypt): Decrypt ciphertext (to a plaintext file) using a Cloud Key Management Service key.
- **`gcloud logging logs list`** (/sdk/gcloud/reference/logging/logs/list): List your project's logs.
- **`gcloud sql backups describe`** (/sdk/gcloud/reference/sql/backups/describe): Display info about a Cloud SQL instance backup.
- **`gcloud sql export sql`** (/sdk/gcloud/reference/sql/export/sql): Export data from a Cloud SQL instance to a SQL file.

Introductory primer

A quick primer for getting started with the gcloud CLI.

Installing the Google Cloud CLI

Install the Google Cloud CLI with these [installation instructions](/sdk/docs/install) (/sdk/docs/install).

Flags, arguments, and other wondrous additions

Arguments can be positional arguments or flags:

- **Positional args:** Set after command name; must respect order of positional args.
- **Flags:** Set after positional args; order of flags doesn't matter.

A flag can be either a:

- *Name-value pair* (`--foo=bar`), or
- *Boolean* (`--force/no-force`).

Additionally, flags can either be:

- *Required*
- *Optional:* If an optional flag is not defined, the default value is used



Global flags

Some flags are available throughout the gcloud CLI experience, like:

- `--help` (/sdk/gcloud/reference#--help): For when in doubt; display detailed help for a command.
- `--project` (/sdk/gcloud/reference#--project): If using a project other than the current one.
- `--quiet` (/sdk/gcloud/reference#--quiet): Disabling interactive prompting (and applying default values for inputs).
- `--verbosity` (/sdk/gcloud/reference#--verbosity): Can set verbosity levels at `debug`, `info`, `warning`, `error`, `critical`, and `none`.

- `--version` (/sdk/gcloud/reference#--version): Display gcloud version information.
- `--format` (/sdk/gcloud/reference#--format): Set output format as `config`, `csv`, `default`, `diff`, `disable`, `flattened`, `get`, `json`, `list`, `multi`, `none`, `object`, `table`, `text`, `value`, or `yaml`.

Cleaning up results

Get the most from your output with the `filter` (/sdk/gcloud/reference/topic/filters), `format` (/sdk/gcloud/reference/topic/formats), `limit`, and `sort-by` flags.

For Compute Engine instances with prefix `us` and not machine type `f1-micro`:

```
gcloud compute instances list --filter="zone ~ ^us AND -machineType:f1-micro"
```

For a list of projects created on or after 15 January 2018, sorted from oldest to newest, presented as a table with project number, project id and creation time columns with dates times in local timezone:



```
gcloud projects list --format="table(projectNumber,projectId,createTime.date(tz=
--filter="createTime>=2018-01-15T12:00:00" --sort-by=createTime
```

For a list of ten Compute Engine instances with a label `my-label` (of any value):

```
gcloud compute instances list --filter="labels.my-label:*" --limit=10
```

Understanding commands

The underlying patterns for gcloud CLI commands; to aid self-discovery of commands.

Finding gcloud CLI commands

The gcloud CLI is a tree; non-leaf nodes are command groups and leaf nodes are commands. (Also, tab completion works for commands and resources!)

Most gcloud commands follow the following format:

gcloud + release level (optional) + component + entity + operation + positional

For example: gcloud + compute + instances + create + example-instance-1 + --zone=us-central1-a

Release level

Release Level refers to the command's release status.

Example: alpha for alpha commands, beta for beta commands, no release level needed for GA commands.

Component

Component refers to the different Google Cloud services.

Example: compute for Compute Engine, app for App Engine, etc.

Entity

Entity refers to the plural form of an element or collection of elements under a component.

Example: disks, firewalls, images, instances, regions, zones for compute

Operation

Operation refers to the imperative verb form of the operation to be performed on the entity.

Example: Common operations are describe, list, create/update, delete/clear, import, export, copy, remove, add, reset, restart, restore, run, and deploy.

Positional args

Positional args refer to the required, order-specific arguments needed to execute the command.

Example: `<INSTANCE_NAMES>` is the required positional argument for `gcloud compute instances create`.

Flags

Flags refer to the additional arguments, `--flag-name(=value)`, passed in to the command after positional args.

Example: `--machine-type=<MACHINE_TYPE>` and `--preemptible` are optional flags for `gcloud compute instances create`.

Except as otherwise noted, the content of this page is licensed under the [Creative Commons Attribution 4.0 License](https://creativecommons.org/licenses/by/4.0/) (<https://creativecommons.org/licenses/by/4.0/>), and code samples are licensed under the [Apache 2.0 License](https://www.apache.org/licenses/LICENSE-2.0) (<https://www.apache.org/licenses/LICENSE-2.0>). For details, see the [Google Developers Site Policies](https://developers.google.com/site-policies) (<https://developers.google.com/site-policies>). Java is a registered trademark of Oracle and/or its affiliates.



Last updated 2025-10-28 UTC.